

IMAGINARY ELECTRIC COMPANY

Power Generation Division - Plant Integration

Alpilla 600 MW Combined Cycle Power Plant - 1 x IE-9H.02 multi-shaft power island

Steam Turbine IE-ST3R - Datasheet

Document number	IEC-ALP-DS-103
Revision	A
Date	2026-12-04
Status	Revised for EPC engineering - supersedes Rev 0
Addressee	Istanbul EPC Muhendislik ve Taahhut A.S. (consortium leader) - EPC engineering
Contract	ALP-EPC-001 (Owner contract); consortium agreement IEPC-IEC

	Prepared	Checked	Approved
Function	IEC plant integration engineer	IEC lead engineer (discipline)	IEC project engineering manager
Signature / date	signed 2026-12-04	signed 2026-12-04	signed 2026-12-04

Rev	Date	Description
0	2026-09-27	Issued for EPC engineering (reference data)
0	2026-09-27	Issued for EPC engineering
A	2026-12-04	TQ-IEC-025: steam conditions per heat balance Rev B, bypass valve tags

CASE STUDY - Imaginary Electric, its products and all data are fictional.

1. General

Item	Data
Model	IE-ST3R reheat condensing steam turbine, 3,000 rpm, two casings: HP barrel turbine and combined IP/LP turbine with double-flow LP section, side exhaust (both sides) to the condenser
Last stage blade / exhaust annulus	1,070 mm, 12 % Cr steel with erosion shields / 2 x 10.8 m2
Output at SRC-NG-100	200.73 MW at the generator terminals
Maximum continuous output	235 MW (cold ambient and valves wide open)
Section efficiencies (SRC, isentropic, shaft)	HP 89.0 %, IP 92.5 %, LP 88.5 % (to the ELEC, incl. moisture)
Exhaust loss at SRC	24.0 kJ/kg
Governing	throttle (sliding pressure above 50 % load), HP/IP combined stop-control valves, LP admission valve
Bypass valves (supplied by IEC, installed in EPC piping)	HP bypass 100 % to cold reheat; IP bypass 100 % and LP bypass 100 % to condenser dump devices; tags 10LBA10AA501 (HP bypass), 10LBB10AA501 (IP bypass), 10LBD10AA501 (LP bypass)
Gland steam	self-sealing above 40 % load; start-up sealing steam 4.5 t/h at 10 bar(a), 250 - 300 degC from the EPC auxiliary boiler; gland steam condenser with 2 x 100 % exhausters (IEC)
Lube / control oil	ISO VG 46 lube oil, tank 32 m3, 2 x 100 % AC + DC pump; fire-resistant EHC fluid (ER-07.06)
Turning gear	hydraulic, 5 rpm

2. Steam conditions (at the ST terminal points)

Admission	Flow kg/s	Pressure bar(a)	Temperature degC	Design pressure / temperature
HP main stop valve inlet (IF-06)	100.15	181.5	598.5	200 bar(a) / 605 degC
HP exhaust / cold reheat (IF-07)	100.15	40.2	366.2	52 bar(a) / 420 degC
IP stop-control valve inlet (IF-10)	117.49	36.2	598.5	48 bar(a) / 605 degC
LP admission valve inlet (IF-12)	11.90	5.05	249.0	8 bar(a) / 330 degC
LP exhaust (to condenser)	129.39	0.0344	26.4	full vacuum / 1.5 bar(a)

3. Operating limits

Parameter	Value
Exhaust pressure: continuous / alarm / trip	120 / 150 / 250 mbar(a)
Main and reheat steam temperature: continuous maximum / 400 h per year	605 degC / 615 degC
Start-up (ST roll to full load) after 8 h / 48 h / cold	25 / 60 / 150 min
Allowable nozzle loads	per IEC-ALP-IF-001 attachment A (NEMA SM-23 x 3.0 for HP/IP, x 1.5 for LP admission)

4. Weights

Component	Weight (t)	Remark
HP turbine (shipped assembled)	92	heaviest lift in the turbine hall for maintenance: IP/LP rotor 86 t
IP/LP outer casing lower half (2 pieces)	2 x 78	
IP/LP rotor	86	turbine hall crane: 100 t main hook minimum
ST generator stator	185	erection by hydraulic jacking/skidding, not by crane